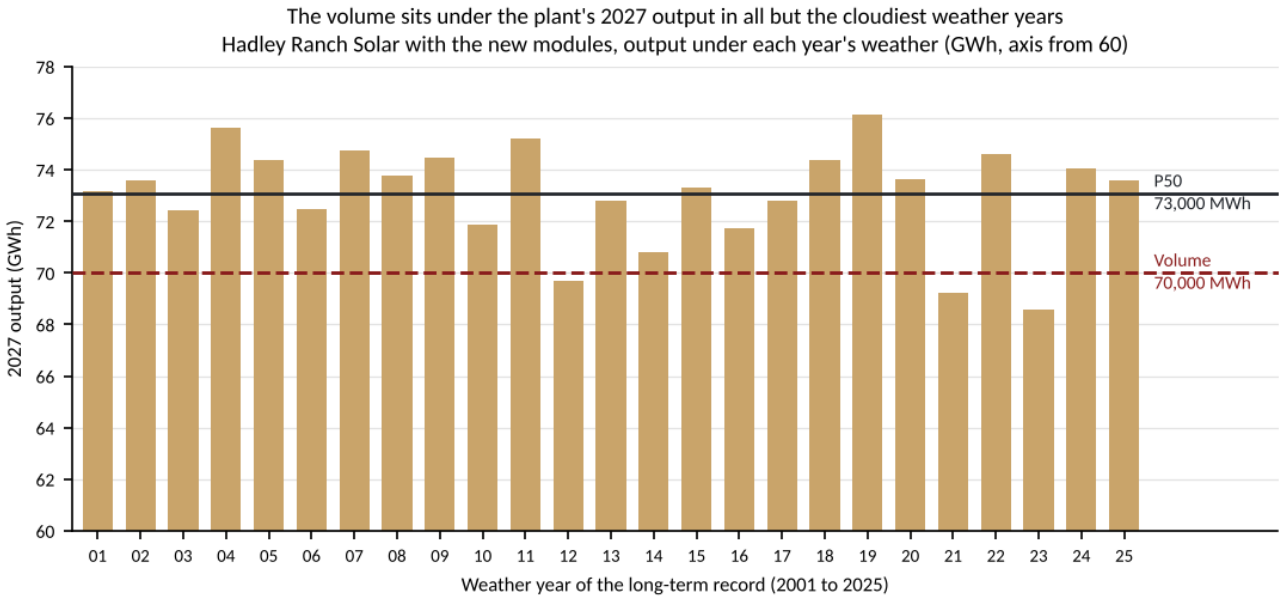
To	Caroline Barnett, Treasurer
From	Kristopher Lara, Energy and Commodities Manager
Date	October 8, 2026
Subject	Hadley Ranch Solar: 2027 forward sale volume for approval (energy risk policy, section 5.1)

Sell 70,000 MWh of Hadley Ranch's 2027 output forward

I recommend we sell 70,000 MWh forward by October 31, the plant's one-year P90 for 2027 with its new modules, rounded down. Across the 25 weather years of Clearfield's record the plant's 2027 output has a P50 of 73,000 MWh and a standard deviation of 1,900 MWh.

The volume starts from what the plant delivered in the twelve months to September 30, a performance ratio of 63.7 per cent, and changes only what the replacement changes. The original modules' own wear goes with them: measured at the inverters' DC input on the days after each block's wash, where the dust and the inverters drop out, it comes to 14 per cent since 2016. The new modules then add the 14.7 per cent per kWp that Corvane measured against unexposed original modules in block 3 over the same twelve months. That puts the repowered plant at a performance ratio of 85 per cent on 37,324.8 kWp.

What the replacement does not touch stays in the number: the inverters, which have run on refurbished power stages since 2018 and are out more often than they were, and the dust that builds between the single December wash the O&M agreement provides through 2027. Carrying the plant's whole decline since 2016 back into 2027 would overstate the sale; carrying today's output without the wear would understate it.



Base period: October 1, 2025 to September 30, 2026, the twelve complete months before the sale (policy section 4.4). P50 and standard deviation over the full calendar years 2001 to 2025 of the site-adapted record (section 4.2); the P90 is the P50 less 1.2816 standard deviations, rounded down to the nearest 1,000 MWh (section 4.1).

Workbook: hedge_volume_2027.xlsx, with the build, the weather years, each inverter's 2025 availability under Schedule 4 and the 2016 to 2025 hedge record.